



Discover and Learn with Captain Cosmic!

Meet Emma Cora

- **Age:** 28
- **Occupation:** Teacher (Primary School Science)
- **Status:** Single
- **Bio:** Emma is a full-time teacher at a primary school in Leicester. She loves going outdoors and being sociable. Her favourite place to visit in Leicester is the National Space Centre.

Goals: Teach her students about Outer Planets and help the class get good grades.

Frustrations: Her students tend to lose focus quickly and forget things.

What does she wish for? Emma hopes for something memorable and educational for her class.

How can this product help Emma? This product can provide a fun way to teach her students about the Outer Planets.

Meet Joey Chester

- **Age:** 10
- **Occupation:** Student (Primary School)
- **Status:** Single
- **Bio:** Joey is an energetic kid who attends a primary school in Leicester. His favourite subject is Science because he loves space and rocket ships. He likes playing video games like Destiny and classics like Space Invaders.

Goals: He wants to learn more about space and rocket ships, participate in science competitions, and improve his grades.

Frustrations: He doesn't like large pieces of text as it tends to bore him.

What does he wish for? Joey hopes for a fun experience that provides interesting space facts.

How can this product help Joey? It can provide a humorous and interactive way to teach Joey about space without large amounts of text.

Empowering The Next Generation Through Laughs and Facts!

Emma wants to find something that the kids will remember and **learn easily** as this will allow for the kids to get good grades.

However, the class is easily distracted and loses focus. Now that she is moving onto a new topic ‘**Outer Planets**’ she is looking for a **new experience** that can **teach the class**.

Joey wants to be smart and have lots of **facts about space and science**. He wants to join science competitions to show other people how good his knowledge is.

However, he finds his classes boring even when it’s about a topic he enjoys. He needs something **without too much text** that is **interactive** and **engaging**.

Emma Needs	Joey Needs	Does my product fulfil their needs (Yes/No)
Learn easily	Facts about space and science	Yes
Outer planets	Without too much text	Yes
New experience	Interactive	Yes
Teach the class	Engaging	Yes

Captain Cosmic’s Rocketship Journey:

The mixture of educational facts, interactive features and its humorous nature will most definitely grab the class's attention and help them learn much better.

This will not only benefit Emma but also Joey as he gets to expand his knowledge in a way that he enjoys.

Competitive Advantage

Our business competitive advantage:

The main business competitive advantage we have with this product is the strong inclusion of an interactive artificial intelligence model.

What makes it unique?:

The fact that we are making our game both interactive and educational is what makes our product stand out when compared to competitors.

New technologies and low costs:

Using new technologies such as ChatGPT will allow us to implement a smart model with a variety of facts and jokes providing endless fun for the users. Additionally, these technologies are mostly free which allows our project to be low cost.

Our team:

Furthermore, our team is highly passionate and dedicated to this project. We have a structured process as we complete Business and Scrum reports weekly. Each member also has a role in the team, Product Owner, Scrum Master, Developer and Designer. Furthermore, we are using the Taiga software to track our development progress which allows us to meet targets on time.

National Space Centre access:

Our access to the National Space Centre allows us to conduct research on how our target audience interacts with the games at the centre. We can ask them questions such as “What is your biggest frustration when playing a game?” and then use this knowledge to make our product better.

MVP Description (Main Menu)



When users press 'CONTROLS'. It will load a scene that consists of a canvas with text explaining the controls. For example, Forward = W.

START

When users press 'START'. The next scene is loaded.

CONTROLS

When users press 'EXIT'. The game will close and will be ready to be started by the next user.

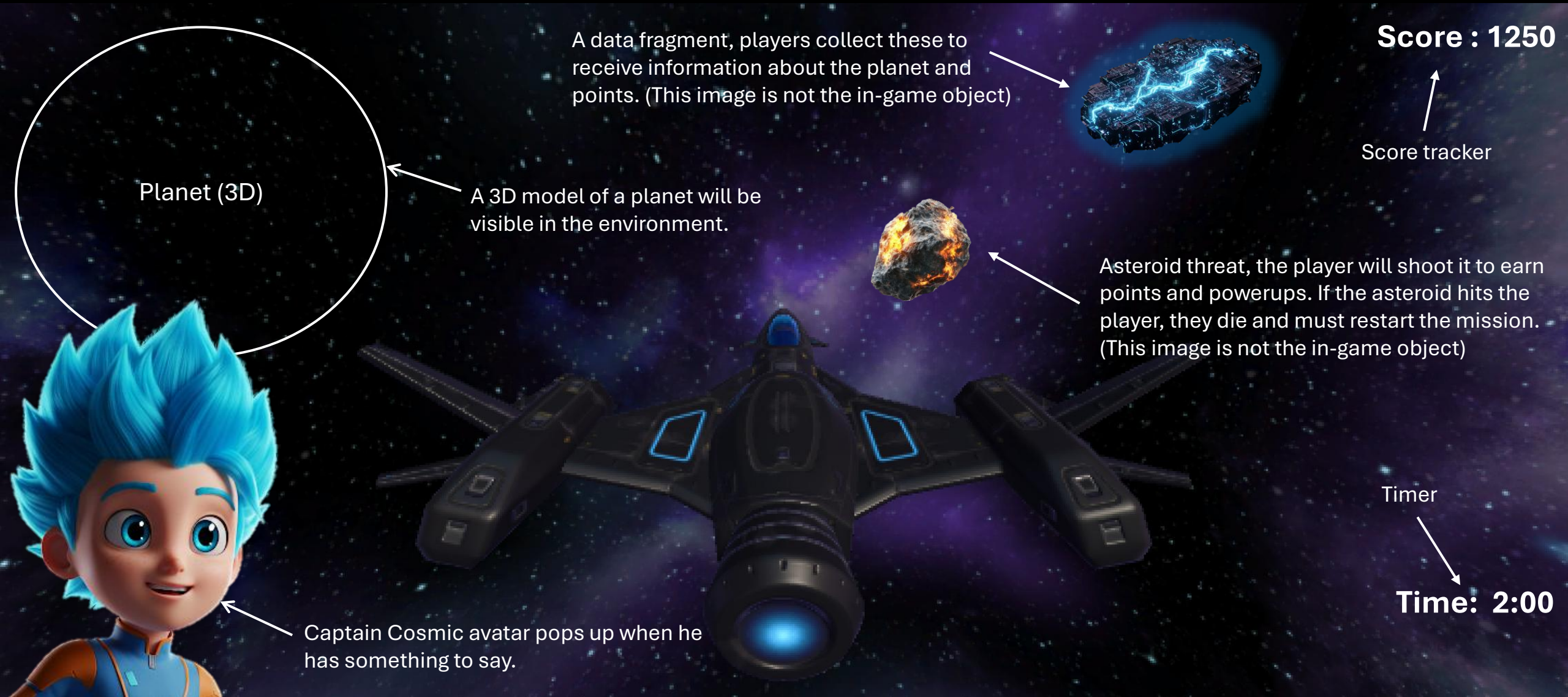
CREDITS

When users press 'CREDITS'. A different scene will be loaded. It will include a canvas with text crediting other developers for the usage of third-party assets.

EXIT



MVP Description (In-game)



A data fragment, players collect these to receive information about the planet and points. (This image is not the in-game object)

Score : 1250

Score tracker

Planet (3D)

A 3D model of a planet will be visible in the environment.

Asteroid threat, the player will shoot it to earn points and powerups. If the asteroid hits the player, they die and must restart the mission. (This image is not the in-game object)

Timer

Time: 2:00

Captain Cosmic avatar pops up when he has something to say.

Captain Cosmic: Welcome to Saturn—the planet with the ultimate bling! Those rings? Made of ice, rock, and dust. Fancy, huh?

Business Model

Product sales related to Captain Cosmic's Rocketship Journey (£)

Market Size

How many potential customers will use my product?

The National Space Centre has 300,000 visitors a year with over 80,000 school children on educational visits (our target audience)

Similar products:

The National Space Centre has many educational space games that are targeted at the same audience. Examples include The Great Rocket Challenge, Life in Space and the Mars Rover Challenge. However, none of them feature a unique avatar like Captain Cosmic.

Different types of users:

- School Children (35%) – Target audience, highest percentage (Educational visits and family visits)
- Teenage students (20%) – Teenage students are likely to play as they could still learn something new.
- Teachers (20%) – Teachers could test the games before teaching the class.
- Parents (10%) – Parents might want to play with their kids on family visits.
- Casual/Other (10%) – The casual visitors might want to try it.
- University Students (5%) – University students are not likely to play, but some might.

Our Business Model

How are we going to monetise?

Our idea is to work with the National Space Centre and an established toy manufacturer to make 'Captain Cosmic' figures. This will be a great way to make the main character more recognisable and generate revenue.

Retail Partner (National Space Centre):

Stocks and sells the figure as part of their gift shop.



Manufacturer (NDA):

Creates the figures and ships them.



Captain Cosmic Figure

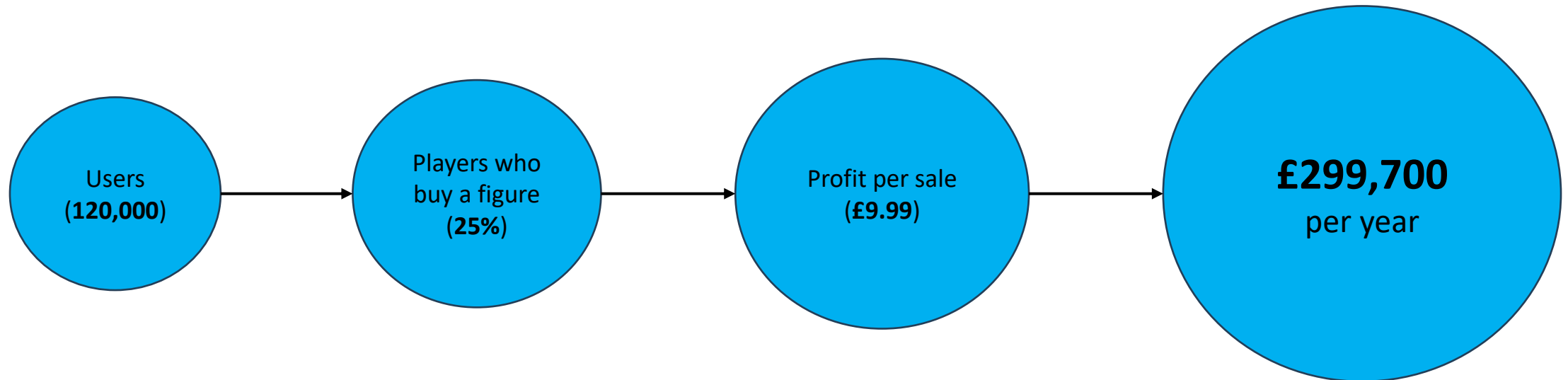
- Price: £12.99
- Production: £3
- Profit: £9.99

Monetisation

Local

National Space Centre:

- 300,000 visitors a year.
- If 40% of visitors play at least once during their visit, we will have 120,000 users a year. (Free to play)
- If 25% of our player base buy a figure, we will sell 30,000 figures locally every year.

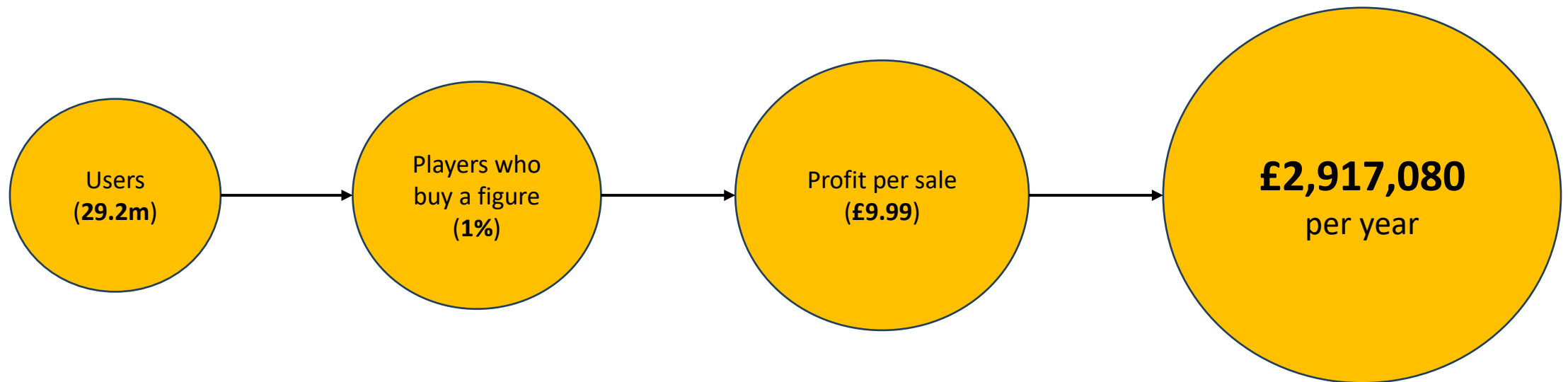


Monetisation

National

United Kingdom:

- 146 million people go to museums/attractions in the UK every year (according to the Association of Leader Visitor Attractions)
- If 20% played at least once during their visit to a museum/attraction, we would have 29.2 million users a year. (Free to play)
- If 1% of our player base buy a figure, we will sell 292,000 nationally every year.

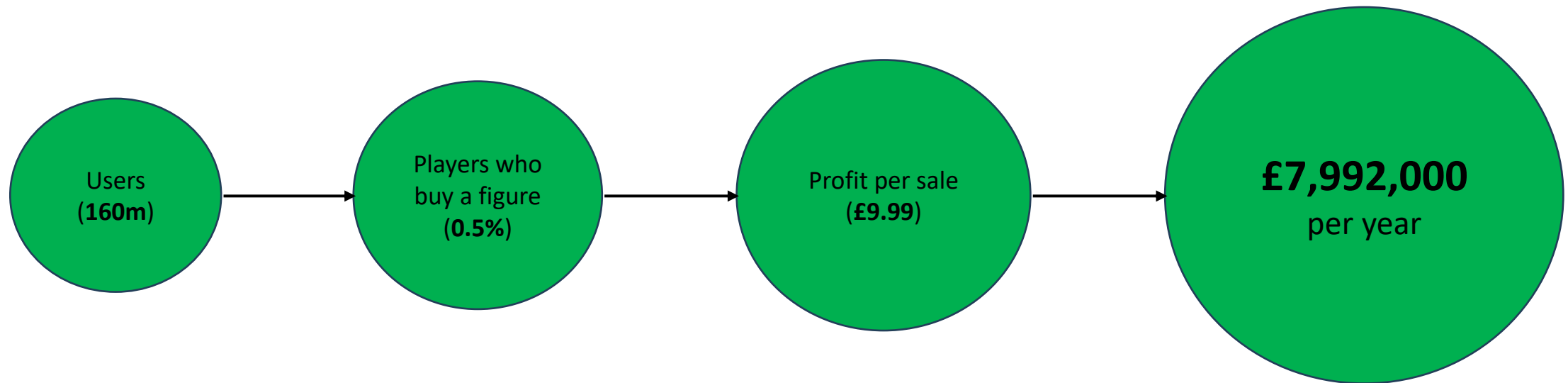


Monetisation

International

International:

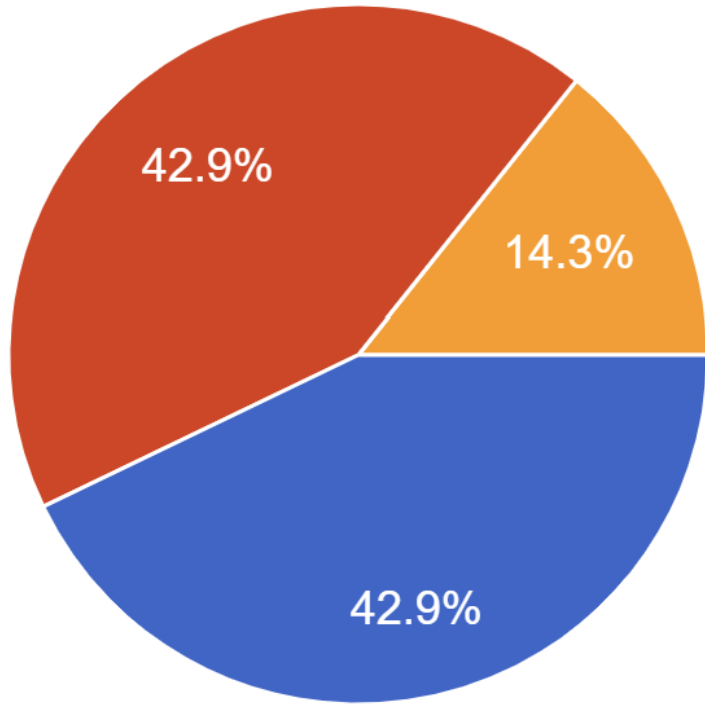
- 1.6 billion people visit museums/attractions every year (according to a rough estimate generated by Gemini using various real and reliable sources)
- If 10% played at least once during their visit to a museum/attraction, we would have 160 million users a year. (Free to play)
- If 0.5% of our player base buy a figure, we will sell 800,000 figures internationally every year.



Understanding our customers

Finding who our customers are, what they are looking for and what their biggest frustrations are

How do you like learning new information in a game?



- Storytelling
- Interactive tasks/challenges
- Trivia questions

What we learnt from this:

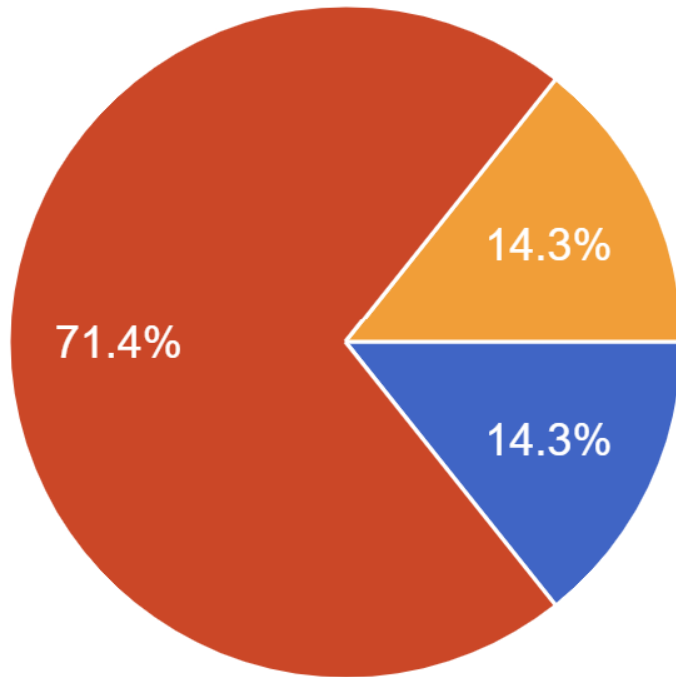
The findings reveal an equal preference between storytelling and interactive tasks/challenges, with trivia questions being the least favoured.

As a result, our game will include a combination of storytelling and interactive tasks/challenge, as their equal significance makes it crucial to include both.

However, we will exclude trivia questions entirely, as they do not appeal to our player base.

Note: We had an uneven number of votes, this led to the overall percentage not exactly adding up to 100%

What is your biggest frustration when playing a game?



- Overload of text/information
- Frequent interruptions
- Confusing controls

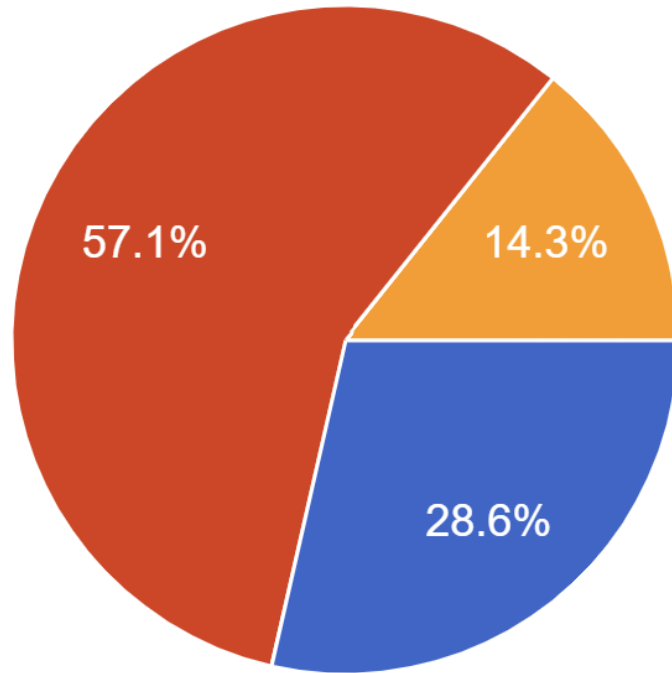
What we learnt from this:

The result indicates that the biggest frustration players experience is frequent interruptions, while overloads of text/information and confusing controls are less frustrating but still notable.

With this insight, we will prioritise on creating a seamless gameplay experience with minimal interruptions.

While overloads of text/information and confusing controls are less critical, they will still be carefully addressed to ensure they don't negatively impact the player experience.

What would you consider a 'fair' price for merchandise?



- £5 - £10
- £10 - £20
- £20 - £30

What we learnt from this:

From this question, we discovered that users consider the £10 - £20 price range as the fairest option compared to £5 - £10 and £20 - £30.

We found these results particularly interesting because we initially assumed that the lowest price range would receive the highest votes. However, the mid-range gathered the best response. This suggests that users are willing to spend a bit more for higher-quality products, but they also have a limit and are not comfortable with prices that are too high. This is evident because the high price range got the lowest votes.

With this insight, we know that our figure price at £12.99 is reasonable, with potential to increase.

Competitors

Who are we competing with?

Competition

Educational, non-space



Educational, space



Merchandising, non-museum



Space Racers Shop



Mars Horizon: Blast Off! (Board game)

Merchandising, museum



Closest competitors

				
Website URL	https://scratch.mit.edu/	https://spaceplace.nasa.gov/	https://www.spaceracers.com/	https://www.kennedyspacecenter.com/
Number of clients / Visitors per month	Website: 21.67M (January 2025)	Website: 875,804 (January 2025)	Website: 2,990 (January 2025)	Website: 213,398 (January 2025) Museum visits: 108,000 per month (estimated)
Service/Product Cost	Free	Free	Game: Free Merchandise: \$19.99	Museum (General Admission): \$24.95 - \$39.95 Merchandise (Gift shop): Prices are unknown.
Strengths	No cost, User-friendly, educational, very popular.	No cost, Reliable and accurate information, educational, supported by NASA.	Engaging, vibrant visuals.	Real-life activity, interactive, educational.
Weaknesses	Doesn't properly teach text-based programming, limitations on what you can create.	The website lacks a balance between education and entertainment, with some games being too focused on one or the other.	Small community, lacks detail.	High admission costs, only accessible from location.
Similarities to your product	Free-to-play, educational.	Free-to-play, educational, space-themed.	3D, educational, space-themed. Sells merchandise.	Educational, space-themed. Sells merchandise.
Differences to your product	Non-space, programming, creating applications.	Multiple games in one application, large amounts of information.	They have different types of media available (Videos)	Real-life activity, NOT free-to-play.

Financial Forecast

What are our incomes and costs?

Company structure and assumptions

- **Number of co-founders:** 4
- **Number of interns:** 2 (Digital Marketing Intern and Accounting Intern)
- **Physical office:** We will rent a small office for planning, meetings, inventory management and development.
- **Organic Marketing:** We can promote it organically through the National Space Centre website, social media and/or events. Furthermore, we can connect with schools and STEM programs to get our target audience to try our game.
- **Manufacturing:** We will use the manufacturing company NDA to make the figures (£3 each)
- **Number of employees:** 6 (Co-founders and Interns only, keeps costs low.)
- **Expected growth rate:** Double our income for each of the first 6 months and then increase our income by 10% for each of the last 6 months. We want to break even after 6 months.
- **Expected number of clients over the 1st year:** 120,000 users (estimated from local monetisation section)

Expenses for the 1st year:

[illegible]

Expected outcome for the 1st year:



Summary

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total	Average
Income	\$895	\$1,790	\$3,580	\$7,160	\$14,320	\$28,640	\$31,504	\$34,654	\$38,119	\$41,931	\$46,124	\$50,736	\$299,454	\$24,955
Expenses	\$14,900	\$14,900	\$14,900	\$14,900	\$14,900	\$14,900	\$14,900	\$14,900	\$14,900	\$14,900	\$14,900	\$14,900	\$178,800	\$14,900
Net savings	-\$14,005	-\$13,110	-\$11,320	-\$7,740	-\$580	\$13,740	\$16,604	\$19,754	\$23,219	\$27,031	\$31,224	\$35,836	\$120,654	\$10,055
Ending balance	\$185,995	\$172,885	\$161,565	\$153,825	\$153,245	\$166,985	\$183,589	\$203,343	\$226,563	\$253,594	\$284,818	\$320,654	\$320,654	\$205,588

Marketing Strategy

How are we going to reach our customers?

Marketing Strategy:

